

Signal Processing and Control Systems (ED213)

A) Grading Policy

Class Attendance (Both Theory and Lab Classes) = 15 %

1st In-Sem= 15 %

2nd In-Sem- 15 %

Course Project= 15 %

Lab Exam =10 %

Endsem =30 %

B) Textbook: I will be following a world-class textbooks in the entire course

Alan V. Oppenheim, A. S. Wilsky, S. H. Nawab, "Signals and Systems," Second Edition, PHI.

Benjamin C. Kuo, Farid Golnaraghi, "Automatic Control Systems," 8th Edition, John Wiley and Sons, 2003.

K. Ogata, Modern Control Engineering, " 4th Edition, Pearson Education, 2002.

C) Exam Schedule and Syllabus w.r.t Course Coverage

- 1) 1st In-Sem -> Chapter 1 (Introduction to Signals and Systems) and Chapter 2 (LTI Systems)
- 2) 2nd In-Sem -> Chapter 3 (Fourier Series), Chapter 4 (Fourier Transform),
- 3) Lab Assignments= MATLAB-based Laboratory Exercises based on course coverage across the semester
- 4) EndSem (Entire Syllabus) = More emphasis on => Chapter 6 (Time and Frequency Characterization, Group Delay Function, 1st and 2nd order systems, etc, Chapter 9 (Laplace Transform), and Chapter 10 (Z-transform) and Chapter 11 on Feedback Control Systems

Note:

- Fundamental knowledge of first four chapters is important perform well in Endsem,
- Chapter numbers above refer to the Oppenheim's textbook mentioned above.

D) Attendance and Regularity:

As per attached lecture plan, the theory and lab classes are highly coherent (and not just series of seminars!) and hence, you are hereby asked to be very regular in both theory and lab classes. However, in case of unavoidable circumstances and you are not coming to class, as a courtesy and an elementary discipline, please drop me an email about your absence,

Course Abstract:- The main objective of this course is to conduct a study on fundamental aspects of signals and systems. This course is the gateway to many important courses at undergraduate and postgraduate level such as **Analog and Digital Communication Systems (in fact there cannot be any communication without fundamental knowledge of this course)**, **Digital Signal Processing (DSP)**, **Advanced DSP**, **Digital Image Processing**, **VLSI Signal Processing**, **Machine Learning and Deep Learning**, **Radar Signal Processing**, **Acoustics**, **Speech Communication**, **Seismology**, **Circuit Design**, **Control System Engineering**, **Robotics**, chemical process control, , etc. In fact, the scope of potential and actual applications of the methods of signals and system analysis continues to expand as engineers are confronted with the new challenges involving the synthesis or analysis of complex processes. For these reasons a course in signals and systems not only is an essential element in an engineering program but also can be of the most rewarding, exciting, and useful courses that engineering student take during their engineering education.

The course is designed by keeping in mind to develop in parallel the methods of analysis of continuous-time and discrete-time signals and systems so that students can use computers to develop and implement signals and systems algorithm. This approach also offers a distinct and extremely important pedagogical advantage. Specifically, we are able to draw on the similarities between continuous-time and discrete-time signals and systems methods in order to share insights and intuition developed in each domain. Similarly, we can exploit the difference between them to sharpen an understanding of the distinct properties of each.

Course Outline: - The course first builds up fundamentals and introduction of signals and systems ranging from their types, properties and different examples. The course then discusses extremely useful class of systems, viz., linear time-invariant (LTI) systems and their representation with the help of convolution sum (which can used to model many physical processes). Fourier analysis ranging from Fourier series, Fourier transform, time and frequency characterization is discussed in a greater depth followed by study on Shannon's sampling theory. The course then shifts to Z-transform and Laplace transform with detailed discussion on properties and their applications. The course also discusses in detail design of digital resonators and their applications in addition to discussion on application of signals and systems methods to control systems and communication systems.

Course Instructor

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Class Room Instruction Plan for Signals and Systems

Lecture No.	Details of the lecture
1	Introduction and motivation for signals and systems and their potential applications in engineering field. Exposure to the students for different potential applications of signals and systems in real life environments. Four distinct problems addressed in signals and systems
2	Mathematical background-I: Complex number system and understanding importance of $j = \sqrt{-1}$. Understanding concept of linear combination of vectors. Mathematical background-II: Derivation of Euler's relation $e^{j\theta} = \cos(\theta) + j\sin(\theta)$ and other associated corollaries. Concept and motivation for inner product
3	Introduction to discrete-time signals and systems: Signals, its different types and classifications. Concept and motivation behind signal energy and average power. Standard test signals: Impulse, step, ramp, real and complex exponential.
4	Periodic and aperiodic signals. Difference between periodicity property of continuous-time and discrete-time signals. Basic algebraic operations on signals and illustration of <i>sifting property</i> of signals
5	Introduction to systems I: Examples of systems, modeling of different systems such as R-C circuits or automobile systems by first order differential and difference equation and hence understanding real goal and motivation behind SAS.
6	Properties of systems such as causality, stability, inverse systems along with different examples. Linearity, time-invariance, systems with or without memory along with different examples.
7	Introduction to special class of systems: Linear Time-Invariant (LTI) systems and impulse response of an LTI system.
8	Development of convolution sum and its properties (commutative, cascade, association, parallel) Conceptual Graphical and analytical method of convolution.
9	Problem solving on convolution for finite/infinite duration sequence, etc.
10	Causality and stability conditions for LTI systems in terms of its impulse response.
11	Concept and motivation behind defining autocorrelation and cross-correlation functions, problems solving.
12	Linear Constant Coefficient Difference Equation and its link with LTI system properties and auxiliary conditions. Introduction to Finite Duration Impulse Response (FIR) and Infinite Duration Impulse Response (IIR) systems with examples.
13	Singularity Analysis: understanding impulse response characterizes an LTI system completely, area under impulse is 1, sampling property of impulse, doublet and k^{th} order derivative and integration of impulse.
14	Fourier Analysis: Introduction and motivation- applications. Concept of eigenvalue and eigenvector with respect to LTI system theory and Fourier Analysis and their link with solution of differential equations which represents a mathematical model of LTI system, Importance of sinusoidal signal excitation in LTI system theory
15	Fourier Analysis (contd.): Concept of inner product and its interpretation in terms of

	linear combination. Trigonometric Fourier series representation through orthogonal functions
16	Fourier Analysis (contd.): Exponential Fourier series representation for pulse train signal and conclusion there from. The Dirichlet conditions for convergence of Fourier series Michelson's effect. Fourier Series and LTI systems
17	Fourier Analysis (contd.): Developments of continuous-time Fourier transform (CTFT) from limiting process on Fourier series representation . FT interpretation: Graphical and in terms of inner product
18	Fourier Analysis (contd.): Problem solving for CTFT (especially CTFT for rectangular pulse, sinc function, periodic signal, impulse train, etc.) and hence understanding <i>Heisenberg's uncertainty principle</i> in signal processing framework.
19	Fourier Analysis (contd.): Properties of FT (e.g., shifting, scaling, convolution, multiplication, duality, <i>Parseval's</i> theorem, etc.) Fourier Analysis (contd.): The Dirichlet conditions for convergence of Fourier Transform. Concept and motivation behind negative frequency.
20	Fourier Analysis (contd.): Development of concept of amplitude modulation from modulation theorem Understanding motivation behind defining energy spectral density (ESD), average power and power spectral density (PSD).
21	Fourier Analysis (contd.): Development of Hilbert Transform and its applications in communications systems such as single sideband modulation (SSB)
22	Fourier Analysis (contd.): Understanding LTI filtering using Fourier analysis
23	Fourier Analysis (contd.): Motivation for time and frequency characterization of signals and systems , Magnitude-phase representation of Fourier transform,
24	Fourier Analysis (contd.): Magnitude-phase representation of frequency response of the systems, effect of phase in signal reconstruction. Concept of linear phase and its effect on filtered signal.
25	Fourier Analysis (contd.): Understanding group delay concept, phenomenon of pulse dispersion due to nonconstant group delay, using group delay to understand impulse response of an LTI system, motivation for using decibel (dB) and hence Bode plot for analog system function.
26	Laplace transform: Introduction and motivation, definition of unilateral and bilateral Laplace transform, Examples with Region of Convergence
27	Laplace transform (contd.): Geometric Evaluation of Fourier transform from pole-zero plot, Properties of Laplace transform
28	Laplace transform (contd.): Inverse Laplace transform with examples. Analysis of transient in electric circuits and solution of linear differential equations
29	Motivation for working with polynomials in engineering design, the link between Weirstrass-Stone Approximation theorem and system function of an LTI system.
30	Z-transform: Introduction and motivation. Z-transform of some elementary signals Concept of Region of Convergence (ROC). ROC for finite duration and infinite duration (left and /or right handed) sequences, properties of ROC.
31	Z-transform (contd.): Geometric Evaluation of Fourier transform from pole-zero plot, Inversion of Z-transform: By inspection, power series method, partial-fraction method.
32	Z-transform (contd.): Cauchy's Residue method. Solution of difference equations with the help of Z-transform. Relation between Fourier and Z-transforms. Concept

	of stability of LTI system in Z-domain.
33	Design of Digital Resonator – relationship between -3dB bandwidth and digital pole radius, relationship between resonant frequency and digital pole angle
34	Design of Digital Resonator (Contd): Impulse response of resonator and trade-off between design parameters, and its relevance for design of Biquad Circuits in EVD
35	Feedback Control Systems Open loop vs closed loop systems, Linear Feedback Systems wrt design of Feedback Amplifiers in EVD, Design of Inverse Systems
36	Feedback Control Systems Stabilization of Unstable Systems, Sampled Data Feedback Control Systems, Design of Tracking Systems
37	Feedback Control Systems Root Locus Analysis of Linear Feedback Systems Nyquist Stability Criterion Bode Plots. Gain Margin and Phase Margin
38	Concluding discussion on signals and systems

Representative MATLAB Laboratory Experiments	
Introduction to MATLAB Programming, Writing Functions , File Handling	
Implementation of Convolution Sum	
Implementation of Autocorrelation Function and Cross-Correlation Function	
Significance of Fourier Series Representation	
Computation of Energy Spectral Density (ESD) and Power Spectral Density (PSD)	
Estimation of Instantaneous Frequency (IF) using Hilbert Transform	
Frequency Response of FIR and IIR Filters	
Significance of Phase of Fourier Transform: A case study for speech and image signals	
Implementation of Group Delay Function	